

**Dr. Babasaheb Ambedkar Marathwada University
Chhatrapati Sambhajinagar- 431001**



**Four Year UG Engineering / Technology Programme
(B.E. / B. Tech.)**

**Structure of B.E. / B. Tech. Programme
(Computer Science & Engineering)**

(AS PER NEP-2020)

Effective from 2026-27

Semester –V										
Option 1 : Student can choose option 1 to get B.E./B.Tech with Multidisciplinary Minor (Following structure)										
Sr. No.	Course Category	Course Code	Course Title	Contact Hours		Credits		Scheme of Examination		
				Theor y	Practical	Theor y	Practical	CIA	SEE	Total
1	Program Core Course (PCC)	PCCT-1	Data Base Management System	2	----	2	----	20	30	50
		PCCP-1	Practical Based on Data Base Management System		2		1	20	30	50
		PCCT-2	Software Testing & Quality Assurance	3	----	3	----	40	60	100
		PCCP-2	Practical Based on Software Testing & Quality Assurance		2		1	20	30	50
		PCCT-3	Design Analysis & Algorithm	3	----	3	----	40	60	100
		PCCP-3	Practical Based on Design Analysis & Algorithm		2		1	20	30	50
		PCCT-4	Theory of Computation	3	----	3	----	40	60	100
		PCCP-4	Practical Based on Full Stack Development(Front End)		2		1	20	30	50
2	Programme Elective Courses (PEC)	PECT-1	1 Data Mining & Data Warehousing 2 Digital Image Processing	3	----	3	----	40	60	100
		PECP-1	Practical Based on 1 Data Mining & Data Warehousing 2 Digital Image Processing	---	2	---	1	20	30	50
3	Multidisciplinary Minor (MDM)	MDMT-3	Theory –6 (Choose any one course from the other discipline : Minor Basket)	3	----	3	----	40	60	100
4	Open Elective (OE) to be chosen compulsorily from the faculty other than Major Discipline	OET-4	Theory -7 (Choose any one from the Basket. Respective BoS Group will prepare a Basket)	2	----	2	----	20	30	50
				19	10	19	5	340	510	850
Option 2 : Student can choose option 2 to get B.E./B.Tech Honors with Multidisciplinary Minor (above structure + following additional Honors degree course)										
5	Honors Degree Course	HT-3	Cloud Infrastructure & Services	3	----	3	----	40	60	100
Please note that the course listed under Sr. No. 5 is optional and specifically designed for students pursuing a Bachelor's Degree Honors with Multidisciplinary Minor. Only those students who have secured a minimum of 7.5 CGPA (equivalent to 75% marks) in their first year will be eligible to opt for this option										
Option 3 : Student can choose option 3 to get B.E./B.Tech with Double Minor (Option 1 structure + following additional Double Minor course)										
6	Double Minor Course	DMT-3	Web Programming	3	----	3	----	40	60	100
Please note that the courses listed under Sr. No. 6 is optional and specifically designed for students pursuing a Bachelor's Degree with Double Minor. Only those students who have secured a minimum of 7.5 CGPA (equivalent to 75% marks) in their first year will be eligible to opt for this option.										

Semester –VI

Option 1 : Student can choose option 1 to get B.E./B.Tech with Multidisciplinary Minor (Following structure)

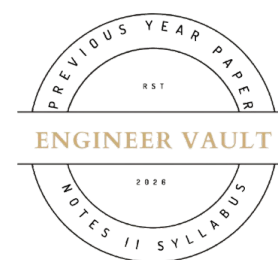
Sr. No.	Course Category	Course Code	Course Title	Contact Hours per week		Credits		Scheme of Examination		
				Theory	Practical	Theory	Practical	CIA	SEE	Total
1	Program Core Course (PCC)	PCCT-1	Artificial Neural Network	3	----	3	----	40	60	100
		PCCP-1	Practical Based on Artificial Neural Network	--	2	---	1	20	30	50
		PCCT-2	Data Science	3	----	3	----	40	60	100
		PCCP-2	Practical Based on Data Science		2		1	20	30	50
		PCCT-3	System Programming & Compiler Construction	3	----	3	----	40	60	100
		PCCP-3	Practical Based on System Programming & Compiler Construction		2		1	20	30	50
2	Programme Elective Courses (PEC)	PECT-1	1) Computer Vision 2) Machine Learning	3	----	3	----	40	60	100
		PECP-1	Practical Based on 1) Computer Vision 2) Machine Learning	--	2	--	1	20	30	50
		PECT-2	1) Cloud Technology 2) Speech & Language Processing	3	----	3	----	40	60	100
		PECP-2	Practical Based on 1) Cloud Technology 2)Speech & Language Processing	--	2	--	1	20	30	50
3	Multidisciplinary Minor (MDM)	MDMT-4	Theory –6 (Choose any one course from the other discipline : Minor Basket)	2	----	2	----	20	30	50
4	Vocational and Skill Enhancement Courses (VSEC)	VSEC-3	Full Stack Development(Back end)	--	4	---	2	20	30	50
5	Research Methodology	RMT	Research Methodology	2	----	2	----	20	30	50
		RP	Research Project-1	--	4	--	2	20	30	50
				19	18	19	9	380	570	950

Option 2 : Student can choose option 2 to get B.E./B.Tech Honors with Multidisciplinary Minor (above structure + following additional Honors degree course)

6	Honors Degree Course	HT-4	Cloud Architecture	3	----	3	----	40	60	100
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Please note that the course listed under Sr. No. 6 is optional and specifically designed for students pursuing a Bachelor's Degree Honors with Multidisciplinary Minor. Only those students who have secured a minimum of 7.5 CGPA (equivalent to 75% marks)

in their first year will be eligible to opt for this option										
Option 3 : Student can choose option 3 to get B.E./B.Tech with Double Minor (Option 1 structure + following additional Double Minor course)										
7	Double Minor Course	DMT-4	Java Programming	3	---	3	---	40	60	100
Please note that the courses listed under Sr. No. 7 is optional and specifically designed for students pursuing a Bachelor's Degree with Double Minor. Only those students who have secured a minimum of 7.5 CGPA (equivalent to 75% marks) in their first year will be eligible to opt for this option.										
Exit Option: Students may choose to exit the program at the end of the third year and be awarded an B. Voc in the relevant discipline or subject, provided they have earned additional 8 credits through skill-based vocational courses, internships, or mini project undertaken during the summer vacation, after third year.										



Total Credits of all Eight Semester

Options	Sem I	Sem II	Sem III	Sem IV	Sem V	Sem VI	Sem VII	Sem VIII	Total Credits
Option 1 (B.E. / B.Tech with Multidisciplinary Minor)	22	22	22	22	24	28	20	16	176
Option 2 (B.E. / B.Tech Honors and Multidisciplinary Minor)	22	22	22 + 3	22 + 3	24+3	28+3	20+3	16+3	194
Option 3 (B.E. / B.Tech Double Minors (Multidisciplinary and Specialization Minor)	22	22	22 + 3	22 + 3	24+3	28+3	20+3	16+3	194
Option 4 (B.E. / B.Tech Honors with Research and Multidisciplinary Minor)	22	22	22	22	24	28	20+9	16+9	194

Course Code : PCCT-1
Course Title : Data Base Management System
(Semester v)

Total Credits : 2

Total Contact Hours : 02Hrs/week ; 30 Hrs / Semester

Maximum Marks : 50

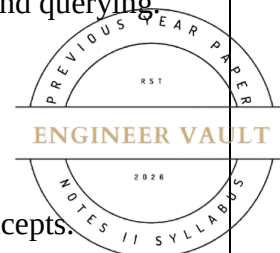
Learning Objectives of the Course:

1. To understand the fundamentals of Database Management Systems and relational database concepts.
2. To develop skills in SQL programming for database creation, manipulation, and querying.
3. To understand relational algebra operations and advanced SQL concepts.
4. To learn PL/SQL programming using procedures, functions, and triggers.
5. To gain basic knowledge of NoSQL databases such as MongoDB.

Course Outcomes (COs) :

After successful completion of the course, students will be able to:

1. Understand DBMS architecture, database models, and relational database concepts.
2. Apply SQL commands for creating, updating, and managing databases.
3. Design ER diagrams and implement relational schemas for real-world applications.
4. Develop SQL queries using joins, aggregate functions, nested queries, and constraints.
5. Implement PL/SQL programs using Procedures, Functions, and Triggers.
6. Demonstrate basic operations of NoSQL databases such as MongoDB



Module No.	Topics / actual contents of the syllabus	Contact Hours
I	A. Introduction to DBMS: Purpose of DBMS, Characteristics of Database Approach, Advantages of DBMS, Data Models, Database Users, Database Architecture, Schema and Instance, Three Level Architecture. B. Relational Database Concepts: Relational Model, Relational Algebra Operations (Select, Project, Join, Union), SQL Basics, DDL, DML, DCL, TCL Commands, Constraints and Keys.	10Hrs
II	A. Database Design and ER Modeling: Entity, Attributes, Relationships, Cardinality, ER Diagram for Library Management and College Admission Systems, Mapping ER Model to Relational Model. B. Normalization Techniques: Functional Dependency, Need for Normalization, First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), Boyce-Codd Normal Form (BCNF), Lossless Join and Dependency Preservation.	10Hrs
III	A. Advanced SQL and PL/SQL: Joins, Subqueries, Aggregate Functions, Group By, Having Clause, Views, Indexes, Procedures, Functions, Triggers, Exception Handling. B. Introduction to NoSQL Databases: Need of NoSQL, Types of NoSQL Databases, MongoDB Architecture, CRUD Operations in MongoDB, Comparison between SQL and NoSQL Databases.	10Hrs

Text Books(If Any):

1. Database System Concepts
2. Fundamentals of Database Systems
3. SQL, PL/SQL The Programming Language of Oracle

Reference Books:

1. Database Management Systems
2. Modern Database Management
3. MongoDB: The Definitive Guide

Course Code: PCCP-1

Course: Lab. : Practical Based on Data Base Management System (Semester :-V)

Credits: 1 Practical: 2 Hrs/week ; 30 Hrs / Semester Maximum Marks : 50

Learning Objectives of the Course:

1. To understand the fundamentals of Database Management Systems (DBMS) and relational database concepts.
2. To learn and apply SQL commands including DDL, DML, DCL, and TCL.
3. To design Entity Relationship (ER) diagrams for real-world applications.
4. To understand PL/SQL concepts including Procedures, Functions, and Triggers.
5. To gain basic knowledge of NoSQL databases and MongoDB operations.

Course Outcomes (COs) :

After successful completion of this course, students will be able to:

1. Understand the concepts of DBMS, database architecture, and relational data models.
2. Apply SQL commands to create, modify, retrieve, and manage database objects and records.
3. Develop database applications using joins, aggregate functions, subqueries, and constraints.
4. Implement PL/SQL programs using Procedures, Functions, and Triggers for database automation.
5. Demonstrate the use of NoSQL databases such as MongoDB for handling unstructured data.

Topics / actual contents of the syllabus	Contact Hours
1.To understand and implement relational algebra operations (Select, Project, Join, Union) using SQL queries. 2.To design and draw an E-R diagram for a Library and College Admission system, identifying entities, attributes, and relationships. 3.Install and configure a DBMS (MySQL, Oracle, or PostgreSQL)./ Create account and use basic DDL ,DML commands using https://freesql.com/ 4.To understand and apply normalization techniques by converting an unnormalized table into 1NF, 2NF, and 3NF. 5. Use the following schema to answer all questions Table: Employees (EmpID, Name, Salary, DeptID, ManagerID) Table: Departments (DeptID, DeptName, Location) Table: Projects (ProjectID, ProjectName, Budget) I. Create the Employee, 'Departments' table with DeptID as the Primary Key and Projects . II. Insert five records in each table III. Write a query to display Employee names and their Department names. 6. Write examples for following SQL Commands using 5 experiment table schema , a) SELECT DISTINCT b) IN c) Alter table d) Like e) Group by f) Having	30 Hrs

g) Drop Table

7. Create a database having two tables with the specified fields, to computerize a library system of a University College. LibraryBooks (Accession number, Title, Author, Department, PurchaseDate, Price), IssuedBooks (Accession number, Borrower)

- a) Create the tables and insert at least 5 records in each table.
- b) Delete the record of book titled “Database System Concepts”.
- c) Change the Department of the book titled “Discrete Maths” to “CS”.
- d) List all books that belong to “CS” department.

8. Create the following tables and answer the queries given below:

Customer (CustID, email, Name, Phone, ReferrerID)

Bicycle (BicycleID, DatePurchased, Color, CustID, ModelNo)

BicycleModel(ModelNo, Manufacturer, Style)

Service (StartDate, BicycleID, EndDate)

- i. In Customer table, CustID is the primary key & ReferrerId is a foreign key
- ii. In Bicycle table, BicycleID is the primary key , CustId is a foreign key & ModelNo is also foreign key
- iii. In BicycleModel table, ModelNo is the primary key.
- iv. In Service table, BicycleId is a foreign key.

Note :Create table in this order: (i), (iii), (ii), (iv)

- a) Create the tables and insert at least 5 records in each table.
- b) Display all customers name with descending order.

9. **Write SQL statement for following (Use Exp. 8 Tables)**

- a) List all the customers who have the bicycles manufactured by manufacturer “Honda”.
- b) List the bicycles purchased by the customers who have been referred by Customer “C1”
- c) List the manufacturer of red colored bicycles.
- d) List the models of the bicycles given for service.

10. Write a PL/SQL Program to demonstrate Procedure

11. Write a PL/SQL Program to demonstrate Function

12. Write a PL/SQL program to demonstrate Triggers.

13. Basic Study of NoSQL Software – MongoDB

Students should undertake at least 08 to 10 experiments during the semester from above list

Course Code : PCCT-2
Course Title : Software Testing & Quality Assurance
(Semester v)

Total Credits : 3 **Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester**
Maximum Marks : 100

Learning Objectives of the Course:

1. To understand the fundamental concepts, principles, and importance of software testing and quality assurance.
2. To learn different software testing levels, types, and testing methodologies used in software development.
3. To understand software quality standards, quality models, and software process improvement techniques.
4. To gain practical knowledge of automated testing tools and frameworks used in industry.
5. To analyze and ensure software reliability, performance, security, and maintainability through quality assurance practices.

Course Outcomes (COs) :

1. Understand software testing concepts and testing methodologies.
2. Apply testing techniques for defect detection.
3. Design effective test cases and manage testing activities.
4. Use automation tools for software testing.
5. Analyze software quality assurance practices and standards.

Module No.	Topics / actual contents of the syllabus	Contact Hours
I	A.Introduction to Software Testing Software Engineering Concepts Need for Software Testing Verification and Validation Testing Principles Software Testing Life Cycle (STLC) Levels of Testing: Unit, Integration, System, Acceptance Testing Functional and Non-functional Testing B.Testing Techniques Black Box Testing,White Box Testing,Grey Box Testing,Boundary Value Analysis Equivalence Class Partitioning,Decision Table Testing,Path Testing and Control Flow Testing	15Hrs
II	A.Test Management Test Planning,Test Case Design,Test Data Preparation,Test Execution and Reporting,Defect Life Cycle Bug Tracking Tools Risk-based Testing B.Automation Testing Introduction to Test Automation,Automation Frameworks,Selenium Basics ,Record and Playback Testing,Data-driven and Keyword-driven Frameworks,Continuous Integration and Testing	15Hrs
III	A.Software Quality Assurance	15Hrs

	Quality Concepts and Standards, Software Quality Models, Capability Maturity Model (CMM), ISO Standards, Software, Reviews and Audits, Metrics for Quality Assurance, Reliability and Performance Testing B.Advanced Topics in Testing Agile Testing, DevOps and Testing, Security Testing, Performance and Load Testing Mobile Application Testing, Cloud-based Testing, Emerging Trends in Software Testing	
Text Books(If Any): 1. Software Testing: Principles and Practices – Naresh Chauhan, Oxford University Press. 2. Software Testing – Ron Patton, Sams Publishing. 3. Foundations of Software Testing – Dorothy Graham, Erik Van Veenendaal, Isabel Evans & Rex Black, Cengage Learning.		
Reference Books: 1. Software Engineering – Ian Sommerville, Pearson Education. 2. Effective Methods for Software Testing – William E. Perry, Wiley Publishing. 3. Lessons Learned in Software Testing – Cem Kaner, James Bach & Bret Pettichord, Wiley India. 4. Introduction to Software Testing – Paul Ammann & Jeff Offutt, Cambridge University Press.		

Course Code: PCCP-2

**Course: Lab. : Practical Based on Software Testing & Quality Assurance
(Semester :-V)**

Credits: 1 Practical: 2 Hrs/week ; 30 Hrs / Semester Maximum Marks : 50

Learning Objectives of the Course:

1. Understand software testing processes and testing life cycle activities.
2. Apply various test case design techniques for software validation.
3. Develop and execute test plans, test cases, and defect reports.
4. Use automation testing tools and frameworks effectively.
5. Analyze software quality using performance and security testing techniques.
6. Demonstrate practical skills in software quality assurance and testing methodologies.

Course Outcomes (COs) :

1. Understand fundamental concepts, principles, and methodologies of software testing and quality assurance.
2. Apply black-box, white-box, and automation testing techniques to identify software defects.
3. Design and execute effective test cases, test plans, and defect management processes.
4. Analyze software quality standards and quality assurance practices for reliable software development.
5. Develop practical solutions for improving software reliability, maintainability, and overall quality.

Topics / actual contents of the syllabus	Contact Hours
1. Study of Software Testing Life Cycle (STLC). 2. Design test cases using Equivalence Partitioning technique. 3. Design test cases using Boundary Value Analysis. 4. Perform Black Box Testing on a sample application. 5. Perform White Box Testing using Control Flow Graph. 6. Prepare test plan and test report for a software application. 7. Study and implementation of Unit Testing using JUnit/PyUnit. 8. Perform Integration Testing on a sample module. 9. Defect reporting and bug tracking using bug tracking tools. 10. Automation testing using Selenium WebDriver. 11. Develop Data-Driven Testing framework using Selenium. 12. Perform Performance Testing using JMeter/LoadRunner.	30 Hrs

Students should undertake at least 08 to 10 experiments during the semester from above list

Course Code : PCCT-3
Course Title : Design and Analysis of Algorithms
(Semester v)

Total Credits : 3

Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester

Maximum Marks : 100

Learning Objectives of the Course:

1. Understand fundamental concepts of algorithm design and computational complexity.
2. Analyze the efficiency of algorithms using asymptotic notations.
3. Apply various algorithm design techniques such as Divide and Conquer, Greedy Method, Dynamic Programming, and Backtracking.
4. Design efficient algorithms for solving real-world computational problems.
5. Develop problem-solving skills through the implementation and analysis of algorithms.

Course Outcomes (COs) :

1. Understand the fundamentals of algorithms and analyze their performance using asymptotic notations.
2. Apply divide-and-conquer and greedy strategies to design efficient solutions for computational problems.
3. Design and implement algorithms using dynamic programming and backtracking techniques.
4. Solve optimization and graph-related problems using appropriate algorithmic approaches.
5. Evaluate, implement, and optimize algorithms for practical applications in computing and engineering domains.

Module No.	Topics / actual contents of the syllabus	Contact Hours
I	A. Introduction & Introduction to Algorithms, What is a "good" algorithm? Criteria for analyzing algorithms. Framework for Analysis: Space Complexity and Time Complexity. Asymptotic Notations: Big-O (O), Omega (Ω), and Theta (Θ) notations with mathematical definitions. Worst-case, Best-case, and Average-case analysis of algorithms (e.g., Linear search vs. Binary search). B. Analysis Framework Analysis of Non-Recursive algorithms (Loops, nested loops, conditional statements). Analysis of Recursive algorithms: Setting up recurrence relations. Solving Recurrence Relations: The Substitution Method and Recursion Tree Method.	15Hrs

II	A.Divide-and-Conquer & Greedy Techniques Divide-and-Conquer: General method, master recurrence for divide-and-conquer.Merge Sort: Algorithm, tracing, and time complexity analysis ($O(n \log n)$).Quick Sort: Algorithm, pivot selection strategies (best, worst, average-case analysis).Strassen's Matrix Multiplication vs. Standard Matrix MultiplicationGreedy Method: General characteristics of greedy algorithms vs. optimization problems.Fractional Knapsack Problem: Greedy strategy based on profit-per-weight ratio.Huffman Coding: Data compression technique, tree construction, and prefix codes.Minimum Spanning Trees (MST): Prim's and Kruskal's algorithms B.Dynamic Programming Introduction to Dynamic Programming: Principle of Optimality, Memoization (Top-down) vs. Tabulation (Bottom-up).0/1 Knapsack Problem using Dynamic Programming: State table formulation and complexity analysis.Longest Common Subsequence (LCS): Problem definition, recursive formulation, and DP matrix.Matrix Chain Multiplication: Optimizing the order of parenthesization.Single-Source Shortest Paths: Bellman-Ford Algorithm (handling negative edge weights).All-Pairs Shortest Paths: Floyd-Warshall Algorithm.Optimal Binary Search Trees (OBST).Comparison between Greedy Method and Dynamic Programming (e.g., Fractional vs 0/1 Knapsack).	15Hrs
III	A. Graph Algorithms & Advanced Search Techniques Graph Traversal: Breadth-First Search (BFS) and Depth-First Search (DFS) with time complexities.Topological Sorting Single-Source Shortest Path: Dijkstra's Algorithm (Greedy implementation using priority queues). B.Backtracking: General method, State-Space Tree representation.The N-Queens Problem and Sum of Subsets problem using Backtracking.Graph Coloring Problem and Hamiltonian Cycles.Branch and Bound: General method, FIFO vs. Least-Cost (LC) search.Solving the Traveling Salesperson Problem (TSP) using Branch and Bound.	15Hrs
Text Books(If Any): 1. Ellis Horowitz, Sartaj Sahni & Sanguthevar Rajasekaran – <i>Fundamentals of Computer Algorithms</i> 2. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest & Clifford Stein – <i>Introduction to Algorithms</i> 3. Anany Levitin – <i>Introduction to the Design and Analysis of Algorithms</i> 4. Alfred V. Aho, John E. Hopcroft & Jeffrey D. Ullman – <i>The Design and Analysis of Computer Algorithms</i> Reference Books: 1. "Fundamentals of Computer Algorithms" by Ellis Horowitz, Sartaj Sahni, and Sanguthevar Rajasekaran 2. "Algorithms" by Sanjoy Dasgupta, Christos Papadimitriou, and Umesh Vazirani		

Course Code: PCCP-3

Course: Lab. : Practical Based on Design and Analysis of Algorithms
(Semester :-V)

Credits: 1 Practical: 2 Hrs/week ; 30 Hrs / Semester Maximum Marks : 50

Learning Objectives of the Course:

1. Understand software testing processes and testing life cycle activities.
2. Apply various test case design techniques for software validation.
3. Perform functional and structural testing on software applications.
4. Use automation testing tools and frameworks effectively.
5. Analyze software quality using performance and security testing techniques.
6. Demonstrate practical skills in software quality assurance and testing methodologies.

Course Outcomes (COs) :

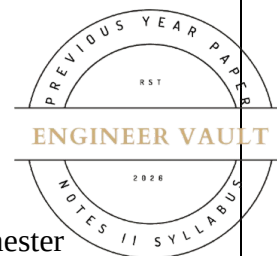
1. Understand fundamental concepts, principles, and methodologies of software testing and quality assurance.
2. Apply black-box, white-box, and automation testing techniques to identify software defects.
3. Design and execute effective test cases, test plans, and defect management processes.
4. Analyze software quality standards and quality assurance practices for reliable software development.
5. Develop practical solutions for improving software reliability, maintainability, and overall quality.

Topics / actual contents of the syllabus	Contact Hours
1. Study of Software Testing Life Cycle (STLC). 2. Design test cases using Equivalence Partitioning technique. 3. Design test cases using Boundary Value Analysis. 4. Perform Black Box Testing on a sample application. 5. Perform White Box Testing using Control Flow Graph. 6. Prepare test plan and test report for a software application. 7. Study and implementation of Unit Testing using JUnit/PyUnit. 8. Perform Integration Testing on a sample module. 9. Defect reporting and bug tracking using bug tracking tools. 10. Automation testing using Selenium WebDriver. 11. Develop Data-Driven Testing framework using Selenium. 12. Perform Performance Testing using JMeter/LoadRunner.	30 Hrs

Students should undertake at least 08 to 10 experiments during the semester from above list

Course Code : PCCT-4

**Course Title : Theory of Computation
(Semester - V)**



Total Credits : 3

Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester

Maximum Marks : 100

Learning Objectives of the Course:

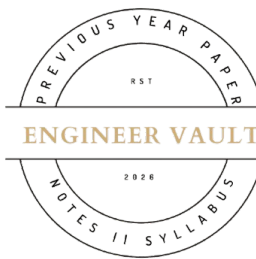
1. Understand the fundamental concepts of formal languages, automata theory and computational models.
2. Analyze finite automata, regular expressions, context free grammars and pushdown automata for language recognition problems.
3. Understand Turing machines, computability and decidability concepts in computational systems.
4. Apply theoretical foundations of computation in compiler design, artificial intelligence and language processing applications.

Course Outcomes (COs) :

1. Explain mathematical constructs and concepts of Theory of Computation.
2. Convert automata, grammar and language representations into equivalent forms.
3. Construct parse trees, CFGs and regular expressions for accepted languages.
4. Analyze regular and context free languages using pumping lemma.
5. Analyze decidability and undecidability of computational problems.
6. Design computational models for language recognition problems.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A. Finite Automata and Formal Languages: Introduction to Theory of Computation, Finite automata, Deterministic Finite Automata (DFA), Non-deterministic Finite Automata (NFA), NFA with ϵ -moves, Conversion of NFA to DFA, Equivalence and minimization of DFA, Finite automata with output and their conversion, Applications of finite automata. B. Regular Expressions and Regular Languages: Regular expressions, Algebraic laws of regular expressions, Finite automata and regular expressions equivalence, Arden's theorem, Applications of Arden's theorem, Pumping lemma for regular languages, Closure properties of regular languages, Applications of regular languages.	15Hrs
II	A. Context Free Grammars and Context Free Languages: Context Free Grammar (CFG), Derivations, Parse trees, Ambiguity in grammar, Simplification of CFG, Elimination of useless symbols, Elimination of null productions, Elimination of unit productions, Chomsky Normal Form (CNF), Greibach Normal Form (GNF), Applications of CFG. B. Pushdown Automata: Pushdown Automata (PDA), Acceptance of PDA, Equivalence of PDA and CFG, Deterministic Pushdown Automata (DPDA), Non-Deterministic Pushdown Automata (NPDA), Pumping lemma for Context Free Languages, Linear Bounded Automata.	15Hrs
III	A. Turing Machines and Computability: Turing machine, Design of Turing machines, Random-access Turing	15Hrs

	machine, Non-deterministic Turing machine, Universal Turing machine, Church's Turing Hypothesis, Halting problem, Unsolvable problems of Turing machine. B. Decidability and Computational Complexity: Decidable problems, Undecidable problems, Recursive languages, Recursively enumerable languages, Post Correspondence Problem (PCP), Computational complexity, Complexity classes P and NP, NP-Complete problems, NP-Hard problems, Tractability and intractability.	
Text Books: 1. Hopcroft J.E., Motwani R., Ullman J.D., <i>Introduction to Automata Theory, Languages and Computation</i>, 3rd Edition, Pearson Education. 2. K.L.P. Mishra and N. Chandrasekaran, <i>Theory of Computer Science – Automata, Languages and Computation</i>, 3rd Edition, PHI Learning.		
Reference Books: 1. John C. Martin, <i>Introduction to Languages and the Theory of Computation</i>, 4th Edition, McGraw Hill. 2. Michael Sipser, <i>Introduction to the Theory of Computation</i>, 3rd Edition, Cengage Learning		

Course Code: PCCP-4 Course: Lab. : Practical Based on Full Stack Development(Front End) (Semester :-V) Credits: 1 Practical: 2 Hrs/week ; 30 Hrs / Semester Maximum Marks : 50	
Learning Objectives of the Course: 1. To develop web pages using HTML5 and CSS3. 2. To design responsive websites using Bootstrap and media queries. 3. To implement JavaScript for client-side validation and interactivity. 4. To develop dynamic front-end applications using ReactJS/Angular basics. 5. To create and deploy responsive web applications. Course Outcomes (COs) : 1. Design static and responsive web pages using HTML and CSS. 2. Implement client-side scripting using JavaScript. 3. Develop responsive layouts using Bootstrap. 4. Create dynamic user interfaces using front-end frameworks. 5. Build and deploy front-end web applications.	
	
Topics / actual contents of the syllabus	Contact Hours
1 Create a basic HTML webpage using headings, paragraphs, lists, and tables. 2 Design HTML forms with various input controls and validations. 3 Create a web page using CSS properties, selectors, margins, and padding. 4 Develop responsive web pages using Bootstrap framework. 5 Create responsive layouts using Flexbox and CSS Grid. 6 Write JavaScript programs using variables, operators, loops, and functions. 7 Implement JavaScript event handling and DOM manipulation. 8 Design a form validation application using JavaScript. 9 Develop a webpage using Fetch API and JSON data handling. 10 Create a mini project using HTML, CSS, and JavaScript. 11 Develop basic ReactJS components and routing. 12 Deploy a responsive front-end project using GitHub Pages.	30 Hrs
Students should undertake at least 08 to 10 experiments during the semester from above list	

Course Code : PECT-1
Course Title : Data Mining & Data Warehousing
(Semester V)

Total Credits : 3

Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester

Maximum Marks : 100

Learning Objectives of the Course:

1. Understand the fundamentals of Data Warehousing and Data Mining concepts.
2. Learn dimensional modeling, ETL processes, and OLAP techniques.
3. Apply preprocessing techniques for improving data quality.
4. Analyze classification, association rule mining, and clustering algorithms.
5. Explore web mining and text mining techniques

Course Outcomes (COs) :

1. Understand Data Warehousing concepts and architecture.
2. Apply preprocessing and transformation techniques.
3. Implement association rule mining and classification techniques.
4. Analyze clustering and web mining techniques.
5. Design data-driven solutions using mining algorithms.
6. Evaluate mining results for decision making.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A. Introduction to Data Warehousing, Data Warehouse Components, Data Warehouse Architecture, Metadata Concepts ,Building a Data Warehouse. B. Dimensional Modeling, Star and Snowflake Schema, ETL Process ,Data Cleaning and Data Quality ,OLAP Operations ,Multidimensional Data Analysis.	15Hrs
II	A. Introduction to Data Mining ,Data Mining Functionalities ,Data Mining Architecture ,KDD Process ,Classification of Data Mining Systems. B. Data Cleaning Techniques ,Data Transformation ,Data Compression ,Dimension Reduction ,Principal Component Analysis ,Binning Methods.	15Hrs
III	A. Association Rule Mining ,Frequent Itemset Mining ,Market Basket Analysis Apriori Algorithm ,Decision Tree Classification ,Bayesian Classification ,Support Vector Machines ,Regression Models. B. Clustering Techniques ,Partitioning Algorithms ,Hierarchical Algorithms Density Based Algorithms ,Web Content Mining ,Web Structure Mining ,Web Usage Mining ,Text Mining and Multimedia Mining.	15Hrs

Text Books(If Any):

1. Data Mining: Concepts and Techniques – Jiawei Han.
2. Data Warehousing, Data Mining and OLAP – Alex Berson.
3. Data Mining Techniques – Arun K. Pujari.
4. Introduction to Data Mining – Pang-Ning Tan.

Reference Books:

1. Data Mining: Introductory and Advanced Topics – Margaret H. Dunham.
2. Data Warehousing Fundamentals – Paulraj Ponniah.
3. Data Mining: Practical Machine Learning Tools – Ian H. Witten.
4. Data Warehousing in the Real World – Sam Anahory.

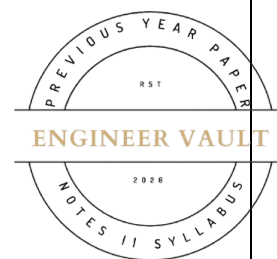
Course Code: PECP-1

Course:DM&DW

(Semester :-V)

Credits: 1 Practical: 2 Hrs/week ; 30 Hrs / Semester

Maximum Marks : 50 / 50



Learning Objectives of the Course:

1. To understand the fundamental concepts of Data Warehousing and Data Mining.
2. To learn dimensional modeling, ETL processes, and OLAP techniques for data analysis.
3. To implement association rule mining, classification, prediction, and clustering algorithms.
4. To explore web mining, text mining, multimedia mining, and real-world analytical applications.
5. To develop analytical, problem-solving, and decision-making skills using mining techniques and business intelligence tools.

Course Outcomes (COs) :

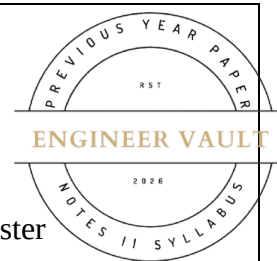
1. Understand the architecture, components, and design principles of Data Warehousing systems.
2. Apply preprocessing, transformation, cleaning, and dimensionality reduction techniques for efficient data preparation.
3. Analyze clustering algorithms and advanced mining techniques including web mining and text mining.
4. Evaluate and interpret mining results and analytical patterns to support effective decision-making in real-world applications.

Topics / actual contents of the syllabus	Contact Hours
1 .Design Star and Snowflake Schemas. 2. Implement ETL operations. 3. Perform OLAP operations. 4. Data preprocessing and cleaning. 5. PCA implementation. 6. Apriori Algorithm implementation. 7. Decision Tree Classification. 8. Clustering using K-Means. 9. Web Mining examples. 10. Text Mining examples. 11. Implementation of K-Means Clustering Algorithm using Sample Dataset. 12. Text Mining and Sentiment Analysis using Real-world Text Data	30 Hrs

Students should undertake at least 08 to 10 experiments during the semester from above list

Course Code : PECT-1

**Course Title : Digital Image Processing
(Semester V)**



Total Credits : 3

Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester

Maximum Marks : 100

Learning Objectives of the Course:

1. To understand the fundamental concepts of digital image processing and image representation techniques.
2. To apply image enhancement, restoration, and filtering techniques for improving image quality.
3. To analyze and implement image transformation, segmentation, and feature extraction methods.
4. To develop image processing solutions using appropriate algorithms, tools, and programming techniques.
5. To evaluate and apply digital image processing techniques in real-world applications such as medical imaging, computer vision, biometrics, and remote sensing.

Course Outcomes (COs) :

1. Explain the fundamentals of digital image processing, image acquisition, sampling, quantization, and pixel relationships.
2. Apply intensity transformation, histogram processing, spatial filtering, and frequency domain techniques for image enhancement.
3. Demonstrate image segmentation techniques such as edge detection, thresholding, region growing, and region splitting and merging.
4. Apply morphological operations and color image processing techniques using different color models and transformations.
5. Analyze image representation and description techniques using boundary and regional description.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A.Introduction to Image Processing Digital image, Fundamental steps in digital image processing, Components of an image processing system, Image sensing and acquisition, Image sampling and quantization, Basic relationship between pixels: Neighbors of a pixel, Adjacency, Connectivity, Regions and Boundaries, Distance Measures. B. Intensity Transformation and Filtering Basics of intensity transformation and filtering, Basic intensity transformation functions, Histogram processing, Histogram equalization, Discrete Fourier transform(DFT), Discrete cosine transform, Fundamentals of spatial filtering, Smoothing spatial filters, Sharpening spatial filters, Basics of filtering in the frequency domain, Image smoothing using frequency domain filters, Image sharpening using frequency domain filters.	15Hrs
II	A.Image Compression Coding Redundancy, Spatial and temporal Redundancy, Irrelevant Information, Measuring image Information, Fidelity Criteria, Image compression Model, Some Basic Compression Methods: Lossless Compression methods-Huffman	15Hrs

	coding, LZW coding, Run- Length Coding, Lossy Compression method: Block Transform Coding, Image File formats: BMP, GIF, TIFF, JPEG, PNG. B.Image Segmentation Point, Line and Edge Detection, Detection of Isolated Points, Line Detection, Edge Models, Basic Edge detection, Thresholding, Region-Based Segmentation Methods: Region Growing, Region Splitting and Merging.	
III	A.Morphological Image Processing and Color Image Processing Morphological Image Processing: Preliminaries, Erosion and Dilation, Opening and Closing The Hit-or-Miss Transformation, Basic Morphological Algorithms: Boundary Extraction, Hole Filling. Color Image Processing: Color Fundamentals and Color Models: RGB color model, CMY and CMYK color models, HSI color model, Basics of Full-Color Image Processing, Color Transformations: Formulation, color complements, color slicing, Tone and color corrections. B.Representation and Description Representation, Boundary Descriptors: Simple descriptors, Shape Numbers, Fourier descriptors, Statistical Moments, Regional Descriptors: Simple descriptors, Topological descriptors.	15Hrs
Text Books(If Any): 1. Rafael Gonzalez and Richard Woods, Digital Image Processing, Pearson Education, 3rd Edition. 2. Rafael C. Gonzalez and Richard E. Woods, Digital Image Processing using MATLAB, Pearson Education, 1st Edition. 3. Anil K. Jain, Digital Image Processing, PHI, 3rd Edition.		
Reference Books: 1. William K. Pratt, Digital Image Processing, McGraw Hill, 3rd Edition. 2. Chanda and D. Dutta Majumder, Digital Image Processing and Analysis, PHI, 2nd Edition.		

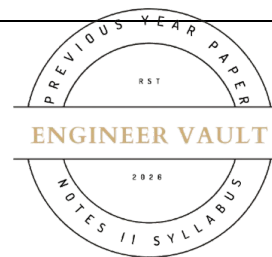
Course: Lab. : PECP-1

Course Code: Digital Image Processing

(Semester :- V)

Credits: 1 Practical: 2 Hrs/week ; 30 Hrs / Semester

Maximum Marks : 50 / 50



Learning Objectives of the Course:

1. To study the Digital Image Processing concepts.
2. To study the different Image processing algorithms to achieve the desired result.
3. To study applications in image processing.

Course Outcomes (COs) :

1. Apply basic image processing and image enhancement techniques including arithmetic operations, intensity transformations, thresholding, histogram processing, smoothing, and sharpening filters.
2. Implement frequency domain and image analysis techniques such as Discrete Cosine Transform (DCT), bit plane slicing, and edge detection.
3. Perform advanced image processing operations including color image processing and morphological operations on digital images.

Topics / actual contents of the syllabus	Contact Hours
1. Write a program to read and display an image. 2. Write a program for arithmetic and logical operations on images. 3. Write a program for image transformations using log transformation, negative transformation, and power law transformation. 4. Write a program to perform thresholding on the image 5. Write a program to perform smoothing (mean, median) filters on an image to reduce noise. 6. Write a program to perform sharpening (Laplacian or high-pass) filters to enhance edges in an image. 7. Write a program for a 1D and 2D discrete cosine transform. 8. Write a program for a histogram and histogram equalization. 9. Write a program for bit plane slicing. 10. Write a program for edge detection. 11. Write a program for color image processing. 12. Write a program for morphological operations on an image.	30 Hrs

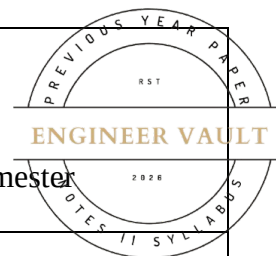
Students should undertake at least 08 to 10 experiments during the semester from above list

Course Code : OET-IV
Course Title : Behavioral Science and organization
(Semester V)

Total Credits : 2

Total Contact Hours : 02Hrs/week ; 30 Hrs / Semester

Maximum Marks : 50



Learning Objectives of the Course:

1. To understand the fundamentals of behavioral science and organizational behavior.
2. To study individual and group behavior in organizations.
3. To develop communication, leadership, and motivational skills.
4. To understand organizational culture, conflict management, and teamwork.
5. To enhance professional and interpersonal effectiveness in workplace environments.

Course Outcomes (COs) :

1. Understand concepts of behavioral science and organizational behavior.
2. Analyze individual behavior, personality, perception, and attitudes in organizations.
3. Apply leadership, communication, and motivational theories in workplace situations.
4. Develop teamwork, conflict resolution, and decision-making abilities.
5. Understand organizational culture and adapt to professional environments effectively.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A.IntroductiontoBehavioral Science Meaning, nature, scope, importance of behavioral science, objectives and applications. B.IndividualBehavior Personality, perception, attitude, values, learning theories, emotional intelligence, stress management.	10Hrs
II	A. Group Behavior and Communication Group dynamics, team building, communication process, barriers to communication, interpersonal communication B.Motivation and Leadership Motivation concepts, Maslow's hierarchy of needs, Herzberg theory, leadership styles, leadership qualities and functions.	10Hrs
III	A. Organizational Behavior Organizational structure, organizational culture, organizational change, conflict management, decision making. B. Professional Ethics and Workplace Skills Professional ethics, time management, workplace behavior, problem-solving skills, adaptability and career development.	10Hrs

Text Books(If Any):

- Stephen P. Robbins – *Organizational Behavior*
- Fred Luthans – *Organizational Behavior*
- K. Aswathappa – *Organizational Behaviour* .

Reference Books:

- Keith Davis – *Human Behavior at Work*
- Udai Pareek – *Understanding Organizational Behaviour*
- Newstrom & Davis – *Organizational Behavior*
- S. P. Robbins – *Essentials of Organizational Behavior*

Course Code : OET-IV
Course Title : Finance & Accounting
(Semester V)

Total Credits : 2

Total Contact Hours : 02Hrs/week ; 30 Hrs / Semester

Maximum Marks : 50

Learning Objectives of the Course:

1. To understand the fundamentals of finance and accounting principles.
2. To develop knowledge of financial statements and accounting procedures.
3. To understand budgeting, costing, and financial management concepts.
4. To analyze financial performance using accounting tools and techniques.
5. To develop practical financial decision-making skills for organizations..

Course Outcomes (COs) :

1. Understand basic concepts of finance and accounting systems.
2. Prepare and interpret financial statements.
3. Apply accounting principles in business transactions.
4. Analyze budgeting, costing, and financial management practices.
5. Develop financial planning and decision-making abilities in organizations.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A. Introduction to Finance and Accounting Meaning, scope, objectives and importance of finance and accounting, accounting principles and concepts, accounting cycle. B. Recording of Transactions Journal entries, ledger posting, trial balance, cash book, bank reconciliation statement.	10Hrs
II	A. Financial Statements Preparation of trading account, profit & loss account and balance sheet. Financial statement analysis and interpretation. B. Cost and Budgetary Control Cost concepts, types of costs, budgeting process, cash budget, budgetary control techniques.	10Hrs
III	A. Financial Management Sources of finance, working capital management, capital budgeting, ratio analysis, break-even analysis. B. Taxation and Financial Planning Basics of taxation, GST concepts, financial planning, investment basics, ethics in finance and accounting.	10Hrs

Text Books(If Any):

- T.S. Reddy & A. Murthy – *Financial Accounting*
- I.M. Pandey – *Financial Management*
- S.N. Maheshwari – *Principles of Accounting*

Reference Books:

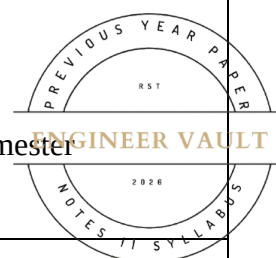
- M.Y. Khan & P.K. Jain – *Financial Management*
- R.K. Sharma & Shashi Gupta – *Management Accounting*
- Charles T. Horngren – *Cost Accounting*
- Jain & Narang – *Accounting Principles*

Course Code : OET-IV
Course Title : Economic Police in India
(Semester V)

Total Credits : 2

Total Contact Hours : 02Hrs/week ; 30 Hrs / Semester

Maximum Marks : 50



Learning Objectives of the Course:

1. To understand the framework of economic policy in India.
2. To study the role of government in economic development.
3. To analyze industrial, agricultural, and trade policies of India.
4. To understand monetary and fiscal policies and their impact on the economy.
5. To develop awareness about current economic reforms and challenges in India.

Course Outcomes (COs) :

After completion of the course, students will be able to:

1. Understand the fundamentals of Indian economic policies.
2. Analyze the role of planning and economic reforms in India.
3. Evaluate industrial, agricultural, and trade policies.
4. Understand fiscal and monetary policy mechanisms.
5. Analyze contemporary economic issues and government initiatives in India.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A. Introduction to Indian Economy and Economic Policies Features of Indian economy, objectives of economic policy, planning in India, NITI Aayog, economic reforms and liberalization. B. Agricultural and Industrial Policies Agricultural policy, Green Revolution, industrial policy resolutions, MSMEs, Make in India initiative.	10Hrs
II	A. Fiscal Policy Meaning and objectives, government budget, taxation policy, public expenditure, public debt, fiscal deficit. B. Monetary Policy Role of Reserve Bank of India, credit control methods, inflation, interest rates, monetary policy tools.	10Hrs
III	A. Trade and Foreign Investment Policies Export-import policy, globalization, foreign direct investment (FDI), balance of payments, WTO and India. B. Contemporary Economic Issues Unemployment, poverty, sustainable development, digital economy, Startup India, Skill India, economic challenges and future prospects.	10Hrs

Text Books(If Any):

- Dutt & Sundaram – *Indian Economy*
- Ramesh Singh – *Indian Economy*
- Mishra & Puri – *Indian Economy*

Reference Books:

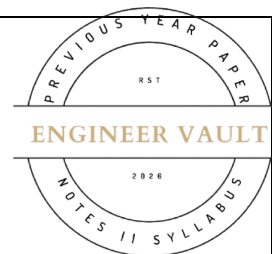
- Uma Kapila – *Indian Economy Since Independence*
- S.K. Agarwal – *Economic Policy of India*
- Government of India – *Economic Survey of India*
- Francis Cherunilam – *Indian Economy*

Course Code : HT-3
Course Title : Cloud Infrastructure & Services
(Semester V)

Total Credits : 3

Total Contact Hours : 03Hrs/week ; 45 Hrs

Maximum Marks : 100



Learning Objectives of the Course:

1. Understand the core concepts, architectures, and service models of cloud computing, including SaaS, PaaS, IaaS, and SECaas.
2. Analyze cloud application design, deployment models, security standards, and performance optimization techniques.
3. Evaluate cloud computing risks, cost models, and data/application security measures for effective cloud adoption.

Course Outcomes (COs) :

1. Understand the fundamental Concepts, architecture and service models of cloud computing.
2. Compare Cloud Computing with Traditional Architectures and Analyze Deployment models like Public, Private and Hybrid.
3. Apply various cloud services such as SaaS, PaaS, IaaS and SECaas to real world applications.
4. Evaluate Cloud Security , Performance, Cost Models and Application Architectures for efficient decision making.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A.Introduction to Cloud Computing : Basic concepts of cloud computing, Characteristics, Layered Architecture and evolution of cloud computing, comparison with traditional computing models and an overview of risks and challenges B. Cloud Types and Deployment Models: Different cloud Types like Public, Private and Hybrid and Their Implementation, Deployment Models and the transition from Traditional to Cloud based Architectures.	15Hrs
II	A.Cloud Service Models: Core service models such as IaaS, PaaS, SaaS, DBaaS and SOA based application, MaaS and CaaS, Case Studies. B.Cloud Security and Risk Management: Introduction of security as a service(SECaas), Cloud security Standards, Risk analysis and Identity Services, Recovery, Storage and Monitoring Services with Risk Mitigation.	15Hrs
III	A. Cloud Applications and Performance Optimization: Cloud Application requirements, Architecture, Client-Server Mechanisms and Performance tuning, Networking issues, automation, self service and parallelization. B.Cloud COsting and Data Management: Cloud Database performance, service bokerage and billing models, Cost Calculation through TCO, Data Management, Data Security.	15Hrs

Text Books and Reference Books (If Any):

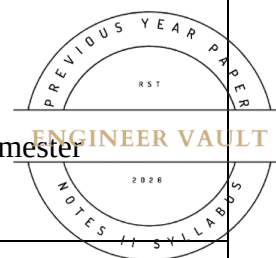
1. Cloud Computing: Concepts, Technology & Architecture - Thomas Erl, Zaigham Mahmood, -Prentice Hall (PEARSON)- 1st Edition.
2. Mastering Cloud Computing: Foundation and Applications Programming. -Rajkumar Buyys,

- Christian Vecchiola- McGrawHill Education- 2nd Edition
3. Cloud Computing Bible- Barrie Sosinsky - Wiley India Pvt.Ltd. - 1st Edition
4. Distributed and Cloud Computing : From Parallel Processing to the Internet of Things. -Kai Hwang, Jack Dongarra - Morgan Kaufmann - 2nd Edition.

Course Code : DMT-3
Course Title : Web Programming
(Semester V)

Total Credits : 3
Maximum Marks : 100

Total Contact Hours : 03 Hrs/week ; 45 Hrs / Semester



Learning Objectives of the Course:

1. To understand the fundamentals of web technologies and internet concepts.
2. To develop static and dynamic web pages using HTML, CSS, and JavaScript.
3. To understand client-side and server-side scripting techniques.
4. To learn database connectivity and web application development concepts.
5. To develop interactive and responsive web applications

Course Outcomes (COs) :

After completion of the course, students will be able to:

1. Understand the architecture and working of web technologies.
2. Design and develop static web pages using HTML and CSS.
3. Implement client-side scripting using JavaScript.
4. Develop dynamic web applications with server-side technologies and database connectivity.
5. Create responsive and user-friendly web applications.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A. Introduction to Web Technologies Internet basics, WWW, web browsers, web servers, client-server architecture, URL, HTTP and HTTPS protocols. B. HTML and CSS Structure of HTML document, text formatting, lists, tables, forms, frames, multimedia elements, introduction to CSS, inline, internal and external style sheets, responsive web design basics.	15Hrs
II	A. JavaScript Introduction to JavaScript, variables, operators, control statements, functions, arrays, events, DOM manipulation, form validation. B. Client-Side Scripting and jQuery JavaScript objects, cookies, jQuery basics, selectors, effects, animations, event handling.	15Hrs
III	A. Server-Side Programming and Database Connectivity Introduction to PHP/Node.js, server-side scripting concepts, GET and POST methods, session and cookies, database connectivity with MySQL. B. Web Application Development CRUD operations, authentication basics, web hosting concepts, introduction to APIs, security issues in web applications.	15Hrs

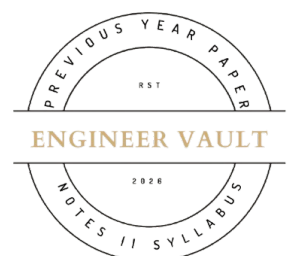
Text Books(If Any):

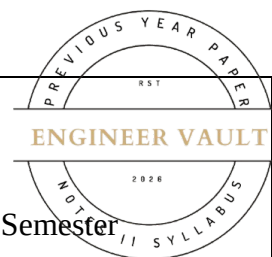
- Jon Duckett – *HTML & CSS: Design and Build Websites*
- Kogent Learning Solutions – *Web Technologies: HTML, JavaScript, PHP, Java, JSP, XML*
- Robin Nixon – *Learning PHP, MySQL & JavaScript*

Reference Books:• Deitel & Deitel – Internet and World Wide Web: How to Program

- Steven Holzner – *JavaScript Complete Reference*
- Thomas Powell – *Web Design: The Complete Reference*
- Jeffrey C. Jackson – *Web Technologies*

Semester VI





Course Code : PCCT-1 Course Title : Artificial Neural Network (Semester VI)	
Total Credits : 3 Maximum Marks : 100	Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester

Learning Objectives of the Course:

1. Understand the fundamental concepts, architecture, and learning mechanisms of Artificial Neural Networks.
2. Analyze different neural network models such as Perceptron, Multilayer Perceptron, and Competitive Networks.
3. Apply training algorithms including Perceptron learning and Backpropagation for solving engineering problems.
4. Design and implement neural network models for pattern recognition, classification, and prediction tasks.
5. Evaluate the performance of various neural network architectures for real-world applications.

Course Outcomes (COs) :

After completion of the course, students will be able to:

- 1: Explain the basic concepts, architecture, and learning paradigms of Artificial Neural Networks.
- 2: Apply Perceptron and Multilayer Perceptron models to solve simple classification problems.
- 3: Implement Backpropagation and competitive learning algorithms for training neural networks.
- 4: Analyze feedback neural networks such as Hopfield and BAM for associative memory applications.
- 5: Design ANN-based solutions for practical applications such as pattern recognition, forecasting, and image processing.

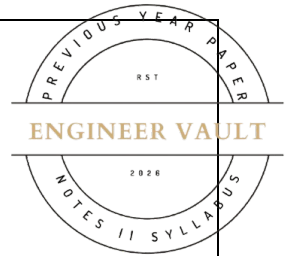
ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A. Characteristics of Neural Networks, Structure and Working of a biological neural network, Artificial Neural Network Terminology, models of neurons: MP model, Perceptron model, Adaline model Topology, Basic Learning laws, what is learning, supervised and unsupervised learning, Functional Units of ANN for pattern, recognition task: Pattern Recognition Problem, Basic functional units. B. Perceptron learning [7 Hrs] Single layer and multilayer perceptron, linear and non-linear separability problems, supervised learning algorithms, Error, correction and Gradient Decent Rules, FFNN, Architecture of, FFNN, Backpropagation learning algorithm, pattern classification,, pattern association by FFNN.	15Hrs
II	A.Associative Learning A.Characteristics of Neural Networks, Structure and Working of a biological neural network, Artificial Neural Network Terminology, models of neurons: MP model, Perceptron model, Adaline model, Topology, Basic Learning laws, what is learning, supervised and unsupervised learning, Functional Units of ANN for pattern recognition task: Pattern Recognition	15Hrs

	Problem, Basic functional units. B.Perceptron learning [7 Hrs] Single layer and multilayer perceptron, linear and non-linear separability problems, supervised learning algorithms, Error correction and Gradient Decent Rules, FFNN, Architecture of FFNN, Backpropagation learning algorithm, pattern classification, pattern association by FFNN. Basic functional units of ANN for pattern recognition tasks: Pattern association, pattern classification and pattern mapping tasks. B.Introduction, Associative Learning, Hopfield network, Error Performance in Hopfield networks, simulated annealing, Boltzmann machine and Boltzmann learning, State transition diagram and false minima problem, stochastic update	
III	A.Components of CL network, Pattern clustering and feature mapping network, ART networks, Features of ART models, character recognition using ART network. B.Self-Organization Maps (SOM): Two Basic Feature Mapping Models, Self-Organization Map, SOM Algorithm, Properties of Feature Map, Learning Vector Quantization, Adaptive Pattern Classification .	15Hrs
Text Books(If Any): 1. Yegnanarayana, “Artificial Neural Networks”, PHI Publications. 2. S. Rajasekaran, Vijayalakshmi Pai, “Neural Networks, Fuzzy Logic and Genetic algorithms- Synthesis and Applications”, PHI Publications.		
Reference Books: 1. Neural Networks a Comprehensive Foundations, Simon Haykin, PHI edition. 2. Laurene Fausett: Fundamentals of Neural Networks: Architectures, Algorithms & Apps, Pearson, 2004. 3. An introduction to neural networks, Gurney, Kevin, CRC press.		

Course Code: PCCP1

Course: Lab. : Artificial Neural Network

(Semester :- VI)



Credits: 1 Practical: 2 Hrs/week ; 30 hrs

Maximum Marks : 50 / 50

Learning Objectives of the Course:

1. Understand the implementation of basic Artificial Neural Network models using programming tools.
2. Develop neural network models for classification, prediction, and clustering problems.
3. Apply various training algorithms such as Perceptron learning and Backpropagation.
4. Analyze the performance of neural networks using suitable evaluation metrics.
5. Design intelligent systems for real-world applications using ANN techniques.

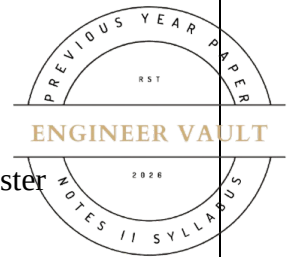
Course Outcomes (COs) :

- 1: Implement basic neuron models and Perceptron learning algorithms.
- 2: Develop Multilayer Perceptron networks using Backpropagation for classification tasks.
- 3: Apply competitive learning and Self-Organizing Maps for clustering problems.
- 4: Design feedback neural networks such as Hopfield Network for associative memory applications.
- 5: Evaluate ANN models using accuracy, error rate etc

Topics / actual contents of the syllabus	Contact Hours
1 . Study and implementation of basic MP model 2. Implementation of Perceptron learning algorithm OR gate 3. Classification of Email into spam and non spam using Perceptron NN 4. Design and training of Backpropagation Neural Network for classification. 5. Performance analysis of ANN by varying learning rate and epochs. 6. Implementation of Hopfield Network for pattern storage and recall. 7. Implementation of Competitive learning Network. 8. Implementation of Kohonen Self-Organizing Map (SOM) for clustering. 9. Handwritten digit recognition using ANN. 10. Prediction of weather using ANN. Note: Practical can be performed using MATLAB / Python	30 Hrs

Students should undertake at least 08 to 10 experiments during the semester from above list

Course Title : Data Science
Course Code : PCCT-2
(Semester VI)



Total Credits : 3

Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester

Maximum Marks : 100

Learning Objectives of the Course:

1. To understand core concepts of Data Science, data preprocessing, and statistical foundations.
2. To apply data analytics lifecycle and build predictive models using Python.
3. To evaluate, visualize data, and understand Big Data technologies for real-world applications.

Course Outcomes (COs) :

- 1: Apply statistical and preprocessing techniques to analyze datasets.
- 2: Develop predictive and classification models using Python and Scikit-learn.
- 3: Evaluate model performance, perform data visualization, and demonstrate understanding of Big Data tools.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction to Data Science: Need, applications, DS vs BI, relationship with Information Science, Data and Preprocessing: Data types, data collection, data wrangling, cleaning, integration, transformation, reduction, discretization, Statistical Foundations: Measures of central tendency and dispersion, correlation, Probability & Inference: Bayes theorem, hypothesis testing, t-test, chi-square test, Case Study: Academic/employee dataset preprocessing and statistical analysis	15Hrs
II	Data Analytics Lifecycle: Discovery, data preparation, model planning, model building, communication, operationalization Python for Data Science: NumPy, Pandas, Matplotlib basics Data Preprocessing in Python: Missing values, duplicates, transformations, Predictive Analytics: Regression: Linear, Logistic, Classification: Naïve Bayes, Decision Trees, Association Rule Mining: Apriori, FP-Growth, Case Study: IRIS dataset – preprocessing and classification	15Hrs
III	Clustering: K-Means, Hierarchical clustering, Model Evaluation: Confusion matrix, precision, recall, F1-score, ROC-AUC, cross-validation, hyperparameter tuning, Time Series Basics Text Analytics: Preprocessing, Bag of Words, TF-IDF, Data Visualization: Line, scatter, histogram, boxplot, density plots Big Data Concepts: Hadoop ecosystem, HDFS, MapReduce, Hive, Pig, Case Study: IRIS dataset clustering and visualization	15Hrs

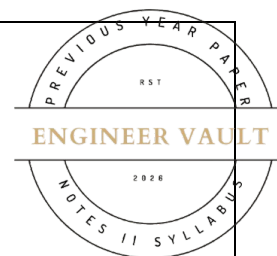
Text Books (If Any):

- J. Han, M. Kamber, and J. Pei, Data Mining: Concepts and Techniques, 3rd ed. Amsterdam, Netherlands: Elsevier, 2011.
- W. McKinney, Python for Data Analysis, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2017.

- D. Dietrich, B. Heller, and B. Yang, Data Science and Big Data Analytics, Indianapolis, IN, USA: Wiley, 2012.

Reference Books:

- J. Han, M. Kamber, and J. Pei, Data Mining: Concepts and Techniques, 3rd ed. Amsterdam, Netherlands: Elsevier, 2011.
- W. McKinney, Python for Data Analysis, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2017.



Learning Objectives of the Course:

1. To develop practical skills in data wrangling, preprocessing, and statistical analysis using Python.
2. To implement predictive analytics, classification, and text analytics techniques on real datasets.
3. To create effective data visualizations and interpret results for data-driven decision making.
4. To gain hands-on exposure to basic machine learning workflows and data analysis tools.

Course Outcomes (COs) :

- 1: Perform data cleaning, wrangling, transformation, and descriptive analysis on datasets.
- 2: Implement regression and classification techniques using Python on real-world datasets.
- 3: Apply text analytics, data visualization, and interpret analytical results effectively.

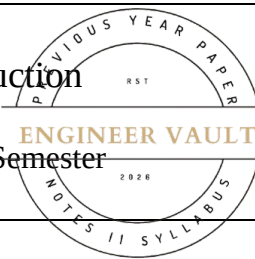
Topics / actual contents of the syllabus	Contact Hours
1 Data Wrangling I: Import open-source dataset, load dataset into pandas DataFrame, inspect data structure, detect missing values, summarize data using describe(), identify variable types, perform type conversion, and encode categorical variables. 2 Data Wrangling II: Create an academic performance dataset and perform missing value treatment, handle inconsistencies, detect and treat outliers, and apply transformation on a selected variable with proper justification. 3 Descriptive Statistics I: Compute mean, median, minimum, maximum, standard deviation, and grouped summary statistics for numeric variables based on a categorical variable. 4 Descriptive Statistics II: Perform statistical analysis of the Iris dataset for different species and compute percentile, mean, and standard deviation with interpretation. 5 Data Analytics I: Implement Linear Regression using Python on the Boston Housing dataset to predict house prices and evaluate model performance. 6 Data Analytics II: Implement Logistic Regression on Social_Network_Ads dataset and compute confusion matrix, accuracy, error rate, precision, and recall. 7 Data Analytics III: Implement Naïve Bayes classification on Iris dataset and compute confusion matrix-based performance measures. 8 Text Analytics I: Perform tokenization, POS tagging, stop-word removal, stemming, and lemmatization on a sample document. 9 Text Analytics II: Construct document representation using Term Frequency and Inverse Document Frequency (TF-IDF).	30 Hrs

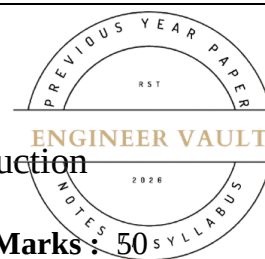
10 Data Visualization I: Use Titanic dataset to explore patterns using Seaborn and plot histogram of fare distribution.

11 Data Visualization II: Plot box plot for age distribution with respect to gender and survival status using Titanic dataset; write observations.

12 Data Visualization III: Analyze Iris dataset, identify feature types, create histograms and boxplots for each feature, and identify outliers.

Students should undertake at least 08 to 10 experiments during the semester from above list

Course Code : PCCT-3 Course Title : System Programming & Compiler Construction (Semester VI) Total Credits : 3 Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester Maximum Marks : 100			
Learning Objectives of the Course: 1. To understand the fundamentals of system software and compiler design. 2. To study assemblers, loaders, macro processors, and linkers. 3. To understand phases of compiler construction. 4. To learn lexical analysis, syntax analysis, and parsing techniques. 5. To develop knowledge of code generation and optimization techniques. Course Outcomes (COs) : After completion of the course, students will be able to: 1. Understand the concepts of system programming and system software. 2. Analyze the working of assemblers, loaders, linkers, and macro processors. 3. Understand different phases of compiler construction. 4. Design lexical analyzers and parsers for programming languages. 5. Apply code generation and optimization techniques in compiler design.			
ModuleNo.	Topics / actual contents of the syllabus	Contact Hours	
I	A. Introduction to System Programming System software, machine structure, language processors, assemblers, loaders, linkers and macro processors. B. Assemblers and Macro Processors Types of assemblers, single pass and two pass assemblers, design of macro processor, macro expansion, conditional macros.	15Hrs	
II	A. Compiler Fundamentals Introduction to compiler construction, phases of compiler, compiler versus interpreter, bootstrapping, error handling. B. Lexical and Syntax Analysis Tokens, lexemes, finite automata, lexical analyzer generation, context free grammar, parsing techniques, top-down and bottom-up parsers.	15Hrs	
III	A. Semantic Analysis and Intermediate Code Generation Syntax directed translation, symbol table management, intermediate code generation, type checking. B. Code Optimization and Code Generation Optimization techniques, loop optimization, peephole optimization, target code generation, storage allocation and error recovery techniques. .	15Hrs	
Text Books(If Any): • Leland L. Beck – <i>System Software: An Introduction to Systems Programming</i> • Alfred V. Aho, Monica S. Lam, Ravi Sethi – <i>Compilers: Principles, Techniques and Tools</i> • D.M. Dhamdhare – <i>Compiler Construction</i>			
Reference Books: 1. Andrew W. Appel – <i>Modern Compiler Implementation</i> 2. Randal E. Bryant – <i>Computer Systems: A Programmer's Perspective</i>			



Course Code: PCCP-3
Course: Lab. : System Programming & Compiler Construction
(Semester :-VI)

Credits: 1 Practical: 2 Hrs/week ; 30 Hrs / Semester

Maximum Marks : 50

Learning Objectives of the Course:

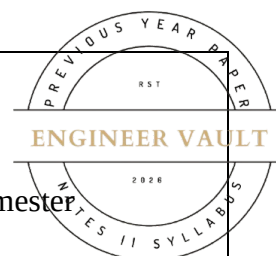
1. To understand practical implementation of system programming concepts.
2. To study assembler, macro processor, loader, and linker operations.
3. To implement lexical and syntax analysis techniques.
4. To develop basic compiler construction programs.
5. To understand code generation and optimization techniques practically.

Course Outcomes (COs) :

1. Implement assembler and macro processor concepts.
2. Design and develop lexical analyzers and parsers.
3. Apply compiler phases in practical applications.
4. Generate intermediate and target code for simple programs.
5. Understand optimization and error handling techniques in compiler construction.

Topics / actual contents of the syllabus	Contact Hours
1. Study of system software tools and compiler phases. 2. Implementation of Pass-I of Two Pass Assembler. 3. Implementation of Pass-II of Two Pass Assembler. 4. Program for Macro Processor implementation. 5. Simulation of Loader and Linker operations. 6. Design of Lexical Analyzer using C/Lex tool. 7. Implementation of Token Recognition using Lexical Analyzer. 8. Construction of Recursive Descent Parser. 9. Implementation of Shift Reduce Parser. 10. Program for Intermediate Code Generation. 11. Implementation of Symbol Table Management. 12. Study and implementation of Code Optimization techniques.	30 Hrs
Students should undertake at least 08 to 10 experiments during the semester from above list	

Course Code : PECT-1
Course Title Computer Vision
(Semester VI)



Total Credits : 3

Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester

Maximum Marks : 100

Learning Objectives of the Course:

1. Understand image processing fundamentals and shape analysis techniques.
2. Learn Hough Transform methods for object detection.
3. Study 3D vision and motion analysis techniques.
4. Explore practical applications of computer vision systems.

Course Outcomes (COs) :

After completion of the course, students will be able to:

1. Explain fundamental image processing techniques used in computer vision systems.
2. Analyze shape analysis methods and implement boundary tracking techniques for object recognition.
3. Apply Hough Transform techniques for detecting lines, circles, and ellipses in digital images.
4. Apply 3D vision and motion analysis techniques for solving computer vision problems.
5. Develop real-world applications using computer vision algorithms and tools.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A. Image Processing Foundations Introduction to Computer Vision, review of image processing techniques, filtering operations, thresholding, edge detection, corner detection, mathematical morphology, texture analysis. B. Shapes and Regions Binary shape analysis, connectedness, object labeling, skeletons, thinning, boundary tracking, active contours, shape recognition, Fourier descriptors, region descriptors and moments.	15Hrs
II	A. Hough Transform Line detection, Hough Transform, line fitting, RANSAC, circular and ellipse detection, generalized Hough Transform, object location, case study: Human Iris Location. B. 3D Vision and Motion Methods for 3D vision, projection schemes, shape from shading, photometric stereo, 3D reconstruction, optical flow, layered motion	15Hrs
III	A. Face Recognition and Surveillance Face detection, face recognition, Eigen faces, surveillance systems, particle filters, tracking and occlusion handling. B. In-Vehicle Vision Systems and Real-Time Applications Roadway detection, road signs identification, pedestrian detection, intelligent transportation systems, industrial applications of computer vision.	15Hrs

Text Books(If Any):

1. D. L. Baggio et al., Mastering OpenCV with Practical Computer Vision Projects, Packt Publishing, 2012.

2. E. R. Davies, Computer & Machine Vision, Fourth Edition, Academic Press, 2012.
3. R. Szeliski, Computer Vision: Algorithms and Applications, Springer, 2011.

Reference Books:

4. Jan Erik Solem, Programming Computer Vision with Python, O'Reilly Media, 2012.
5. Mark Nixon and Alberto S. Aquado, Feature Extraction & Image Processing for Computer Vision, Academic Press, 2012.
6. Simon J. D. Prince, Computer Vision: Models, Learning, and Inference, Cambridge University Press, 2012.

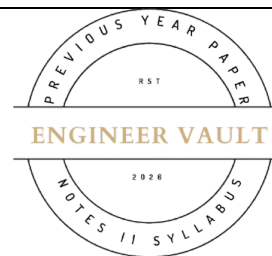
Course Code:

Course: Computer Vision Lab: PECP-1

(Semester :-VI)

Credits: 1 Practical: 2 Hrs/week ; 30 Hrs / Semester

Maximum Marks : 50 / 50



Learning Objectives of the Course:

1. To understand the fundamentals of digital image processing and computer vision techniques.
2. To learn image enhancement, filtering, thresholding, and edge detection methods.
3. To study shape analysis, object recognition, and boundary detection techniques.
4. To understand Hough Transform methods for line, circle, and ellipse detection.
5. To explore 3D vision, motion analysis, and optical flow techniques.
6. To develop basic applications using computer vision and OpenCV tools

Course Outcomes (COs) :

- 1: Implement basic image processing, enhancement, and segmentation techniques using OpenCV tools.
- 2: Apply edge detection, shape analysis, and Hough Transform techniques for object and feature detection.
- 3: Develop simple computer vision applications such as motion detection and face detection using OpenCV.

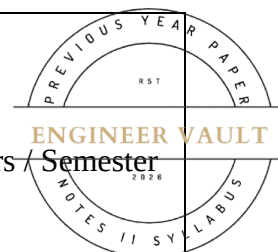
Topics / actual contents of the syllabus	Contact Hours
1 Study and perform basic image processing operations such as image reading, resizing, cropping, and color conversion using OpenCV 2 Implement image smoothing and sharpening techniques for image enhancement 3 Perform image thresholding techniques for image segmentation and binary image creation 4 Implement edge detection techniques using Sobel and Canny edge detectors 5 Detect corners and interest points using Harris Corner Detection method 6 Perform morphological operations such as erosion, dilation, opening, and closing on images 7 Implement object labelling and boundary detection in binary images 8 Perform shape analysis using contour extraction and region properties 9 Detect straight lines in images using Hough Transform technique 10 Detect circles in images using Circular Hough Transform method 11 Implement basic motion detection using optical flow techniques 12 Develop a simple face detection application using OpenCV	30 Hrs
Students should undertake at least 08 to 10 experiments during the semester from above list	

Course Code : PECT-1
Course Title : Machine Learning
(Semester:-VI)

Total Credits : 3

Maximum Marks : 100

Total Contact Hours : 03Hrs/week ; 45 Hrs



Learning Objectives of the Course:

1. To understand fundamental concepts and techniques of Machine Learning.
2. To study supervised and unsupervised learning algorithms.
3. To analyze loss functions, regularization and model evaluation techniques.
4. To explore dimensionality reduction methods and recent trends in Machine Learning.

Course Outcomes (COs) :

1. Explain fundamental concepts and learning paradigms in Machine Learning.
2. Apply supervised and unsupervised learning algorithms for data analysis.
3. Evaluate machine learning models using loss functions and regularization techniques.
4. Analyze dimensionality reduction methods and emerging trends in Machine Learning.
5. Implement basic machine learning models using suitable tools and techniques for simple real-world problem solving.

Module No.	Topics / actual contents of the syllabus	Contact Hours
I	A. Introduction to Machine Learning(7 Hrs):- Introduction to Machine Learning, Comparison of Machine learning with traditional programming, Applications of ML . Machine learning Techniques: Supervised, Unsupervised, reinforcement learning techniques, Real Time Examples, Bias-Variance Tradeoff, Overfitting & Underfitting, Dataset concepts: features, labels, training/testing A. Supervised Learning Techniques (8 Hrs) Regression and Classification -Linear Regression, Logistic Regression, k-Nearest Neighbours (k-NN) , Decision Trees, SVM, Naïve Bayes Classifier, Basic Performance Metrics: Accuracy, Precision and Recall	15Hrs
II	A.Unsupervised Learning Techniques(8 Hrs): Introduction to clustering, Types of Clustering: Hierarchical, Partitional Clustering, k-means clustering , Density based clustering. B. Ensemble Learning and Loss Functions, Regularization(7 Hrs):-Ensemble Learning: Bagging, Boosting, Random Forest Loss Function:-Role of Loss Function, Generalization and regularization, MSE, MAE, Cross-Entropy, Lasso Regression and Ridge Regression for regularization.	15Hrs
III	A. Dimensionality Reduction(9Hrs):- Need of Dimensionality Reduction, feature Selection, Feature Extraction, Dimensionality Reduction Methods, Filter Method, Wrapping Method, Embedded Methods, Applications of Dimensionality Reduction. Techniques Of Dimensionality Reduction:- PCA,CCA, ICA,NMF. B. Recent trends in ML(6 Hrs):-Deep Learning overview, Self	15Hrs

	Supervised Learning, Generative AI, Explainable AI (XAI), Federated Learning, AutoML (Automated Machine Learning).
Text Book	1. Introduction to Machine Learning, By Jeeva Jose, Khanna Book Publishing Co., 2020. 2. Machine Learning for Dummies, By John Paul Mueller and Luca Massaron, For Dummies, 2016. 3. Machine Learning, By Rajeev Chopra, Khanna Book Publishing Co., 2021.
Reference Books:	1. 1. Machine Learning: The New AI, By Ethem Alpaydin, The MIT Press, 2016. 2. 2. Machine Learning, Tom M. Mitchell, McGraw Hill Education, 2017. Online Resources: 1. NPTEL / SWAYAM lectures. 1. https://www.udacity.com/course/intro-to-machine-learning--ud120 2. . https://www.coursera.org/learn/machine-learning-duke

Course Code : PECP-1

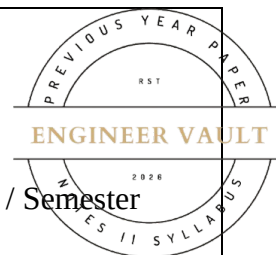
Course: Practical Based on Machine Learning

(Semester :-VI)

Credits: 1

Practical: 2/ Hrs/week ; 30 Hrs / Semester

Maximum Marks : 50



Learning Objectives of the Course:

1. To introduce fundamental concepts, techniques, and applications of Machine Learning.
2. To develop the ability to implement supervised and unsupervised learning algorithms using suitable tools and datasets.
3. To enable students to analyze, evaluate, and apply dimensionality reduction and classification techniques for solving basic real-world problems.

Course Outcomes (COs) :

1. Understand basic concepts and applications of Machine Learning.
2. Implement supervised learning algorithms for prediction and classification.
3. Apply clustering techniques for unsupervised learning problems.
4. Use dimensionality reduction techniques for feature extraction and data analysis.

Topics / actual contents of the syllabus	Contact Hours
1. Study and implementation of basic NumPy and Pandas operations for Machine Learning datasets. 2. Implementation of Simple Linear Regression for prediction on a sample dataset. 3. Implementation of Logistic Regression for classification using Iris/Breast Cancer dataset. 4. Implementation of k-Nearest Neighbors (k-NN) algorithm. 5. Implementation of Decision Tree classifier. 6. Implementation of Support Vector Machine (SVM). 7. Implementation of Naïve Bayes Classifier using Bayes Theorem for classification on a sample dataset. 8. Implementation of K-Means clustering algorithm. 9. Implementation of Hierarchical Clustering using dendrogram analysis. 10. Implementation of Density-Based Clustering (DBSCAN) for cluster identification 11. Implementation of PCA for dimensionality reduction and data visualization. 12. Implementation of Non-Negative Matrix Factorization (NMF) for Dimensionality Reduction and Feature Extraction on a sample dataset.	30 Hrs

Students should undertake at least 08 to 10 experiments during the semester from above list

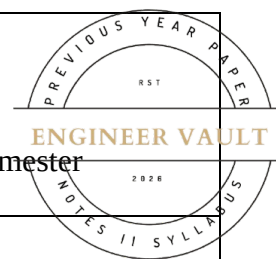
Suggested System Requirements:-

OS:- Windows 10 / 11 or Linux (Ubuntu recommended)

Programming Environment:-Python (Version 3.8 or above)

IDE / Development Tools:-Jupyter Notebook/ Anaconda (*recommended for beginners*)/ Visual Studio Code (*optional*)

Course Code: PECT-2
Course Title: Cloud Technology
(Semester VI)



Total Credits : 3

Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester

Maximum Marks : 100

Learning Objectives of the Course:

1. To understand the fundamentals and architecture of cloud computing.
2. To study cloud service models and deployment models.
3. To learn virtualization and cloud infrastructure technologies.
4. To understand cloud security, resource management, and data storage techniques.
5. To explore modern cloud platforms and container technologies used in industry.

Course Outcomes (COs):

At the end of this course student will:

- 1: Understand the basic concepts, architecture, and service models of cloud computing.
- 2: Analyze virtualization technologies and cloud infrastructure components.
- 3: Implement cloud storage, resource provisioning, and distributed computing concepts.
- 4: Apply cloud security mechanisms and understand cloud performance management.
- 5: Use modern cloud platforms and container orchestration tools for deploying applications.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A. Introduction to Cloud Computing: Definition, Characteristics, Evolution of Cloud Computing, Benefits and Challenges, Distributed Systems, Cluster Computing, Grid Computing, Utility Computing. B. Cloud Service Models and Deployment Models: IaaS, PaaS, SaaS, Public Cloud, Private Cloud, Hybrid Cloud, Community Cloud, Cloud Architecture and Components.	15Hrs
II	A. Virtualization Technology: Introduction to Virtualization, Hypervisors, Types of Virtualization, Server Virtualization, Storage Virtualization, Network Virtualization, Virtual Machine Management. B. Cloud Infrastructure and Resource Management: Data Centers, Cloud Resource Provisioning, Load Balancing, Elasticity, Scalability, Energy Efficient Computing, SLA Management, Deployment de, Cloud Programming and Management, Aneka SDK, Management Tools.	15Hrs
III	A. Cloud Storage and Security: Cloud Storage Architecture, Distributed File Systems, Data Security in Cloud, Identity Management, Authentication and Authorization, Backup and Recovery, Privacy Issues. B. Modern Cloud Platforms and Containers: Introduction to AWS, Microsoft Azure, Google Cloud Platform, Docker Containers, Kubernetes Basics, Container Orchestration, Serverless Computing and Edge Computing	15Hrs

Text Books(If Any):

1. Cloud Computing: Principles and Paradigms
2. Mastering Cloud Computing

Reference Books:

1. Cloud Computing and SOA Convergence in Your Enterprise A Step-by-Step Guide by David S. Linthicum from Pearson 2010.
 2. Cloud Computing 2nd Edition by Dr. Kumar Saurabh from Wiley India 2012.
- Micheal Miller. Pearson Education

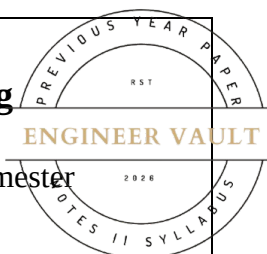
Course Code: PECP-2 Course: Lab. : Cloud Technology (Semester : VI) Credits: 1 Practical: 2 Hrs /week ; 30 Hrs / Semester Maximum Marks : 50 / 50	
Learning Objectives of the Course: 1. To learn the fundamental ideas behind Cloud Computing, the evolution of the paradigm, its applicability. 2. To understand the basic ideas and principles in data center design, cloud management techniques and cloud software deployment considerations. Course Outcomes (COs): 1. Apply cloud computing tools and platforms to develop, deploy, and manage applications and services. 2. Analyze and implement cloud architectures, virtualization, and web service technologies in real-world scenarios.	
Topics / actual contents of the syllabus	Contact Hours
1.Study of Cloud Computing Platforms. 2.Installation and Configuration of Virtual Machine using VirtualBox/VMware. 3.Creation and Management of Virtual Machines. 4.Implementation of Cloud Storage Services. 5.Study and Use of AWS/Azure/GCP Services. 6.Load Balancing and Auto Scaling Demonstration. 7.Cloud Security Configuration and User Access Control. 8.Introduction to Docker and Container Creation. 9. Docker Image Creation and Container Deployment. 10.Kubernetes Cluster Basics and Pod Deployment. 11.Deploying a Web Application on Cloud Platform. 12. Mini Project based on Cloud Application Deployment.	30 Hrs
Students should undertake at least 08 to 10 experiments during the semester from above list	

Course Code: PECT-2
Course Title: Speech & Language Processing
(Semester VI)

Total Credits : 3

Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester

Maximum Marks : 100



Learning Objectives of the Course:

1. To teach the fundamentals of Speech and language processing, and also to make them for understanding CFG, PCFG in NLP.
2. To know the role of semantics of sentences and pragmatic.
3. To teach the basic concepts of speech processing along with analysis and
 - a. modeling.

Course Outcomes (COs) :

After completion of the course, students will be able to -

1. Learn the fundamentals of Speech and language processing
2. Understand the use of CFG and PCFG in Speech and language processing
3. Understand the role of semantics of sentences and pragmatic
4. Introduce Speech Production and Related Parameters of Speech.
5. Show The Computation and use of techniques such as Short Time Fourier
6. Transform, Linear Predictive Coefficients and Other Coefficients in the analysis of speech.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	Introduction Origins and challenges of NLP – language modeling: grammar-based lm, statistical lm –regular expressions, finite-state automata – english morphology, transducers for lexicon and rules, tokenization, detecting and correcting spelling errors, minimum edit distance Word Level Analysis Unsmoothed n-grams, evaluating n-grams, smoothing, interpolation and backoff – word classes, part-of-speech tagging, rule-based, stochastic and transformation-based tagging, issues in pos tagging – hidden markov and maximum entropy models.	15Hrs
II	Syntactic Analysis Context free grammars, grammar rules for English treebanks, normal forms for grammar – dependency grammar – syntactic parsing, ambiguity, dynamic programming parsing – shallow parsing – probabilistic cfg, probabilistic cyk, probabilistic lexicalized cfs – feature structures, unification of feature structures. Semantics And Pragmatics Requirements for representation, first-order logic, description logics – syntax-driven semantic analysis, semantic attachments – word senses, relations between senses, thematic roles, selectional restrictions – word sense disambiguation, wsd using supervised, dictionary & thesaurus, bootstrapping methods – word similarity using thesaurus and distributional methods.	15 Hrs
III	Basic Concepts of Speech Processing Speech fundamentals: articulatory phonetics – production and classification of	15Hrs

	speech sounds; acoustic phonetics – acoustics of speech production; review of digital signal processing concepts; short-time fourier transform, filter- bank and lpc methods. Speech-Analysis,Speech Modeling Features, feature extraction and pattern comparison techniques: speech distortion measures– mathematical and perceptual – log–spectral distance, cepstral distances, weighted cepstral distances and filtering, likelihood distortions, spectral distortion using a warped frequency scale, LPC, PLP and MFCC coefficients, time alignment and normalization – dynamic time warping, multiple time – alignment paths. Hidden markov models: markov processes, HMM – evaluation, optimal state sequence – Viterbi search, Baum-welch parameter re-estimation, implementation issues.	
Text Books(If Any): i) Daniel Jurafsky, James H. Martin—Speech and Language Processing: “An Introduction to Natural Language Processing, Computational Linguistics and Speech”, Pearson Publication, 2014. ii) Steven Bird, Ewan Klein and Edward Loper, —”Natural Language Processing with Python”, First Edition, OReilly Media, 2009. iii) Lawrence Rabiner And Biing-Hwang Juang, “Fundamentals Of Speech Recognition”, Pearson Education, 2003.		
Reference Books: i) Frederick Jelinek, “Statistical Methods Of Speech Recognition”, MIT Press, 1997. ii) Breck Baldwin, —Language Processing with Java and LingPipe Cookbook, Atlantic Publisher, 2015 iii) Richard M Reese, —Natural Language Processing with Java, OReilly Media, 2015.		

Course Code: PECP-2

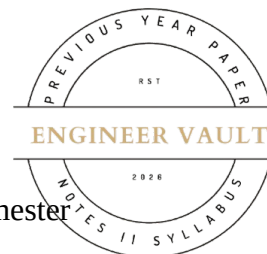
Course: Lab. : Speech & Language Processing

(Semester : VI)

Practical: 2 Hrs/week ; 30 Hrs / Semester

Credits: 1

Maximum Marks : 50 / 50



Learning Objectives of the Course:

1. To teach the python library that suits for text processing. It emphasis on various preprocessing steps to work on text. It also covers real time applications as part of this course.

Course Outcomes (COs) :

After completion of the course, students will be able to -

1. Understand the preprocessing steps for NLP.
2. Apply Python libraries like NLTK for NLP to work on stop words, tokenization, corpus etc.
3. Understand the importance of lemmatization and stemming concepts for NLP.
4. Understand the process of applying NLP for various applications.

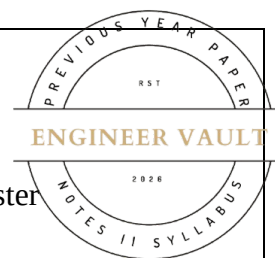
Topics / actual contents of the syllabus	Contact Hours
1. Write a Python NLTK program to remove stop words from a given text. 2. Write a Python NLTK program to split the text sentence/paragraph into a list of words. 3. Write a Python NLTK program to tokenize sentences in languages other than English. 4. Write a Python NLTK program to create a list of words from a given string. 5. Write a Python NLTK program to list down all the corpus names. 6. Write a Python NLTK program to get a list of common stop words in various languages in Python. 7. Write a Python NLTK program to find the definition and examples of a given word using WordNet. 8. Write a Python NLTK program to find the sets of synonyms and antonyms of a given word. 9. Write a Python NLTK program to get the overview of the tagset, details of a specific tag in the tagset and details on several related tagsets, using regular expression. 10. Write a Python NLTK program to find the number of male and female names in the names corpus. Print the first 10 male and female names. 11. Write a Python program to demonstrate Lemmatization with NLTK. 12. Write a Python program to demonstrate Stemming with NLTK. 13. Write a Python Program for NLP analysis of Restaurant review. 14. Write a program for Twitter Sentiment Analysis using Python. 15. Write a program for Sentiment Analysis using VADER.	30 Hrs

Students should undertake at least 08 to 10 experiments during the semester from above list

Course Code: VSEC-3
Course Title: Full Stack Development(Back end)
(Semester VI)

Total Credits : 2
Maximum Marks : 50

Total Contact Hours : 04Hrs/week ; 60 Hrs / Semester



Learning Objectives of the Course:

1. To understand backend architecture and server-side programming concepts.
2. To develop backend applications using modern frameworks and technologies.
3. To implement RESTful APIs and database connectivity.
4. To understand authentication, authorization, and security concepts.
5. To develop scalable and efficient backend services for web applications.

Course Outcomes (COs) :

After completion of the course, students will be able to -

1. Understand backend application architecture and workflows.
2. Develop server-side applications using backend technologies.
3. Implement REST APIs and database operations.
4. Apply authentication and security mechanisms in applications.
5. Build and deploy backend services for full stack applications.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	Frontend Development Introduction to full stack architecture, HTML5, CSS3, Bootstrap, JavaScript basics, responsive web design, form handling and validation. Backend Development using Java Introduction to Java web technologies, Spring Boot architecture, project setup, controllers, REST APIs, dependency injection.	15Hrs
II	Database Connectivity and Application Development MySQL connectivity using Spring Data JPA, entity classes, CRUD operations, exception handling, API testing. Practical: Employee and student record database queries	15 Hrs
III	Authentication and Deployment User authentication, session management, JWT basics, application deployment concepts and mini project development. Right Join processing concepts; short-time fourier transform, filter- bank and lpc methods.	15Hrs

Text Books(If Any):

1. Craig Walls – *Spring Boot in Action*
2. Herbert Schildt – *Java: The Complete Reference*
3. Jon Duckett – *HTML & CSS: Design and Build Websites*
4. Learning SQL – Alan Beaulieu – O'Reilly Latest Edition

Reference Books:

1. Kathy Sierra & Bert Bates – *Head First Java*
2. Spring Documentation – *Spring Boot Reference Guide*
3. Deitel & Deitel – *Internet and World Wide Web: How to Program*

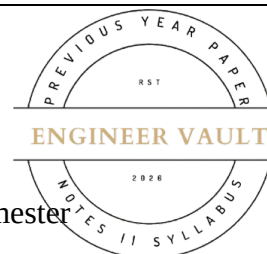
Course Code: VSEC-3

**Course: Lab. : Full Stack Development(Back end)
(Semester : VI)**

Credits: 2

Practical: 4 Hrs/week ; 60 Hrs / Semester

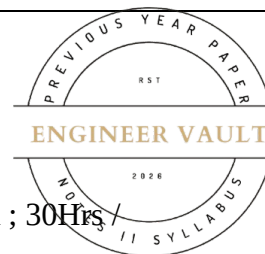
Maximum Marks : 50 / 50



Topics / actual contents of the syllabus		Contact Hours
Sr. No.	Practical Title	60 Hrs
1	Study of Full Stack Development Architecture and Development Environment Setup.	
2	Create a basic HTML5 webpage using headings, lists, tables, and hyperlinks.	
3	Design a registration form using HTML5 form controls.	
4	Develop a responsive webpage using CSS3 styling techniques.	
5	Create a webpage layout using Bootstrap Grid System.	
6	Design responsive navigation bars, cards, and forms using Bootstrap components.	
7	Implement JavaScript programs for variables, operators, and control statements.	
8	Develop JavaScript functions for form validation.	
9	Create dynamic web pages using DOM manipulation and events.	
10	Develop a responsive Student Registration Portal using HTML, CSS, Bootstrap, and JavaScript.	
11	Install and configure Java, Maven, and Spring Boot environment.	
12	Create a basic Spring Boot project using Spring Initializr.	
13	Develop a Spring Boot application with Controllers and Request Mapping.	
14	Implement Dependency Injection using Spring Framework.	
15	Create RESTful APIs using Spring Boot.	
16	Develop REST APIs for Employee Management System.	
17	Design Entity Classes and Repository Interfaces using Spring Data JPA.	
18	Configure MySQL Database Connectivity in Spring Boot.	
19	Perform CRUD Operations on Employee Records using JPA Repository.	
20	Perform CRUD Operations on Student Records using JPA Repository.	
21	Implement Custom Queries using JPQL and Native SQL Queries.	
22	Develop Exception Handling Mechanisms in Spring Boot Applications.	
23	Test REST APIs using Postman or Swagger.	
24	Implement User Registration and Login Module.	
25	Develop Session Management and Access Control Mechanisms.	
26	Implement JWT-based Authentication and Authorization.	
27	Deploy Spring Boot Application on Local/Cloud Server.	
28	Mini Project: Full Stack Employee/Student Management System using HTML, Bootstrap, JavaScript, Spring Boot, MySQL, and JWT Authentication.	
Students should undertake at least 08 to 10 experiments during the semester from above list		

Course Code :RMT

**Course Title : Research Methodology
(Semester VI)**



Total Credits : 2

Total Contact Hours : 02Hrs/week ; 30Hrs

Maximum Marks : 50

Learning Objectives of the Course:

1. To introduce students to the fundamentals of research methodology and scientific investigation.
3. To develop the ability to formulate research problems, review literature, and prepare research proposals.
5. To enable students to understand data collection, analysis techniques, and research ethics.

Course Outcomes (COs) :

- 1: Understand basic concepts, processes, and significance of research methodology.
- 2: Apply appropriate research methods and data collection techniques for problem solving.
- 3: Develop research proposals and interpret research findings with ethical considerations.

Module No.	Topics / actual contents of the syllabus	Contact Hours
I	A. Introduction to Research Methodology: Meaning and Objectives of Research, Research Process, Types of Research, Defining Research Problem. B. Research Design, Literature Review, Research Ethics, Impact Factor, H-Index and Citations.	10Hrs
II	A. Methods of Research: Qualitative and Quantitative Research Methods, Sampling Techniques. B. Questionnaire Design, Interviews, Observation Methods, Data Collection Tools and their Applications.	10Hrs
III	A. Data Analysis Techniques, Statistical Tools, Hypothesis Testing. B. Interpretation of Results, Research Proposal Writing, Technical Paper Writing, Plagiarism and Report Presentation.	10Hrs

Text Books(If Any):

- 1.Kothari C. R., Research Methodology: Methods and Techniques, New Age International.
- 2.Panneerselvam R., Research Methodology, PHI Learning
3. Prabhat Pandey , Research Methodology tools and Techniques , Bridge Center , 2015

Reference Books:

- 1.Creswell J. W., Research Design: Qualitative, Quantitative and Mixed Methods Approaches, SAGE Publications.
2. Merriam S. B. and Tisdell E. J., Qualitative Research: A Guide to Design and Implementation, Wiley.

Course Code :RP

**Course Code: Research Project-1
Semester :-VI**

Credits: 2 Practical: 4 Hrs/week ; 60 Hrs / Semester

Maximum Marks : 50

Learning Objectives of the Course:

1. To enable students to identify and formulate a research problem in emerging areas of Computer Science and Engineering.
2. To develop skills in literature survey, problem analysis, and research methodology.
3. To guide students in designing, implementing, and validating innovative project solutions.
4. To improve technical writing, presentation, teamwork, and project documentation skills.

Course Outcomes (COs) :

- CO1: Identify and analyze research problems in relevant domains.
 CO2: Perform literature survey and select appropriate tools and methodologies.
 CO3: Design and implement a prototype/model for the selected problem.
 CO4: Prepare technical reports and present project progress effectively.

Topics / actual contents of the syllabus	Contact Hours
1. Identification of research problem and project domain selection. 2. Literature survey using journals, conference papers, and online databases. 3. Problem statement formulation and objective definition. 4. Requirement analysis and feasibility study. 5. Selection of tools, platforms, datasets, and technologies. 6. Design of system architecture / methodology / workflow. 7. Dataset collection, preprocessing, and validation techniques. 8. Implementation of preliminary modules / prototype development. 9. Experimental setup and testing procedures. 10. Performance analysis and result interpretation. 11. Preparation of project report, synopsis, and presentation slides. 12. Seminar presentation, project demonstration, and viva-voce.	60 Hrs
Students should undertake at least 08 to 10 experiments during the semester from above list	

Course Code : HT-4
Course Title: Cloud Architecture
(Semester VI)

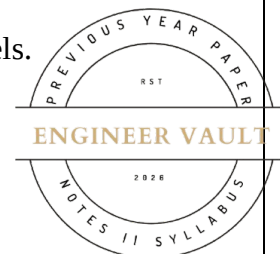
Total Credits : 3

Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester

Maximum Marks : 100

Learning Objectives of the Course:

1. To understand cloud computing fundamentals and cloud architecture models.
2. To study virtualization and containerization technologies.
3. To learn cloud infrastructure, networking, and storage concepts.
4. To understand cloud-native applications and microservices architecture.
5. To study cloud security, reliability, and deployment strategies.



Course Outcomes (COs) :

After completion of the course, students will be able to:

- 1: Understand cloud computing concepts and service models.
- 2: Analyze cloud infrastructure and virtualization technologies.
- 3: Design scalable and reliable cloud architectures.
- 4: Implement containerization and orchestration techniques.
- 5: Apply cloud security and deployment strategies in real-world applications.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A. Introduction to Cloud Computing and Architecture Evolution of cloud computing, characteristics of cloud computing, cloud deployment models, cloud service models (IaaS, PaaS, SaaS), cloud reference architecture, advantages and challenges of cloud computing. B. Virtualization and Containerization Virtual machines, hypervisors, server virtualization, storage virtualization, Docker architecture, containers vs virtual machines, Kubernetes basics and container orchestration.	15Hrs
II	A. Cloud Infrastructure and Networking Cloud data centers, compute services, storage architecture, object storage, block storage, file storage, virtual networking, load balancing, DNS, CDN, software defined networking. B. Cloud Native Applications Monolithic vs microservices architecture, API gateway, RESTful services, serverless computing, event-driven architecture, DevOps and CI/CD pipelines, Infrastructure as Code (IaC).	15 Hrs
III	A. Cloud Security and Reliability Cloud security challenges, IAM, authentication and authorization, encryption techniques, monitoring and logging, disaster recovery, backup strategies, fault tolerance and high availability. B. Cloud Platforms and Emerging Trends Introduction to AWS, Microsoft	15Hrs

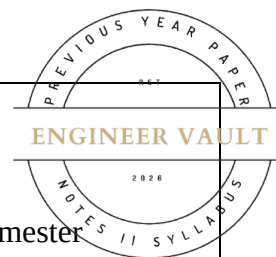
	Azure, Google Cloud Platform, hybrid cloud, multi-cloud architecture, edge computing, green cloud computing, cloud migration and future trends
Text Books(If Any): 1. Richard L. Halterman, “Object-Oriented Programming in Java”, 1st Edition, Southern Adventist University, 2011... 2. Kathy Sierra, Bert Bates, “Head First Java”, 2nd Edition, O'Reilly Media, 2005. 3. Paul Deitel, Harvey Deitel, “Java: How to Program”, 10th Edition, Pearson Education, 2015.	
Reference Books: 1. Kathy Sierra, Bert Bates , “ <i>Head First Java</i> ”, 2nd Edition , O'Reilly Media 2. Herbert Schildt, “Java: A Beginner’s Guide”, 8th Edition, McGraw-Hill Education 3. Bruce Eckel , “Thinking in Java”, 4th Edition, Prentice Hall 4. Joshua Bloch,, “Effective Java”, 3rd Edition, Addison-Wesley 5. Herbert Schildt, “Java: The Complete Reference”, 12th Edition, McGraw-Hill Education	

Course Code :DMT-IV
Course Title: JAVA Programming
(Semester VI)

Total Credits : 3

Total Contact Hours : 03Hrs/week ; 45 Hrs / Semester

Maximum Marks : 100



Learning Objectives of the Course:

1. To learn object oriented concept in Java
2. To study inheritance, polymorphism concept.
3. To understand the exception and multithreading.

Course Outcomes (COs) :

After completion of the course, students will be able to -

1. Describe the fundamental concepts of Java, including OOP principles, JVM architecture, and basic syntax structures like variables, data types, and control statements.
2. Apply object-oriented programming concepts such as classes, objects, constructors, overloading, and access modifiers in Java.
3. Implement inheritance, abstract classes, and method overriding to promote code reuse and polymorphic behavior in Java programs.
4. Design and implement programs using interfaces and packages for modular and extensible software development.
5. Handle errors and exceptions effectively using Java's exception handling mechanisms to build robust programs.

ModuleNo.	Topics / actual contents of the syllabus	Contact Hours
I	A.Introduction to OOP and Java(7 Hrs) Futures of java, difference between procedural oriented and approach oriented, Java Virtual Machine, Data types, Variables, Operators, Control Statements, String & Array in Java. B.OOP in Java (8 Hrs) Class fundamental, introducing method, declaring object, accessing class members, method overloading, constructor, constructor overloading, static member, access modifiers, this references, introducing final.	15Hrs
II	A.Inheritance(7 Hrs) Inheritance in java, super and sub-classes, defining subclass, type of inheritance method overloading, using super, finalizers,abstract class and method, visibility control, using final with inheritance. B.Interfaces & Packages (8Hrs) Interface in Java, Defining interfaces, Extending & Implementing Interfaces, Polymorphism, Packages: Defining a Packages, Class Path Variable, Creation of Package, Importing Packages	15 Hrs
III	A.Exceptional Handling (7 Hrs) Exceptional Handling fundamentals, Java Built in exceptions, Try Catch & Finally, Throws, Throws Keywords, User defined Exception B.Multi-Threading (8Hrs) Definition of Threads, States of Threads,	15Hrs

	Common Threads method, Creation of Threads, Creation of Multiple Threads, Thread priorities, Synchronization
Text Books(If Any): 1. Richard L. Halterman, “Object-Oriented Programming in Java”, 1st Edition, Southern Adventist University, 2011... 2. Kathy Sierra, Bert Bates, “Head First Java”, 2nd Edition, O'Reilly Media, 2005. 3. Paul Deitel, Harvey Deitel, “Java: How to Program”, 10th Edition, Pearson Education, 2015.	
Reference Books: 1. Kathy Sierra, Bert Bates , “ <i>Head First Java</i> ”, 2nd Edition , O'Reilly Media 2. Herbert Schildt, “Java: A Beginner’s Guide”, 8th Edition, McGraw-Hill Education 3. Bruce Eckel , “Thinking in Java”, 4th Edition, Prentice Hall 4. Joshua Bloch,, “Effective Java”, 3rd Edition, Addison-Wesley 5. Herbert Schildt, “Java: The Complete Reference”, 12th Edition, McGraw-Hill Education	

